LEARNER GUIDE

Area of a Circle

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps.
Connection before curriculum. Always.

What this lesson is about

The area of a circle is the amount of flat space inside it. In this lesson you do something better than memorise a formula — you build it. By cutting a circle into sectors and rearranging them into a parallelogram, you discover that the area is $A = \pi r^2$. You will also meet a surprise: because the area depends on r^2, doubling the radius makes the area four times as big. The question you are exploring is: how much space is inside a circle, and how does it depend on the radius?

What you are learning to notice

- That a circle can be rearranged into a parallelogram of base πr and perpendicular height r.
- That the area of a circle is $A = \pi r^2$ — π times the radius squared.
- That the area is proportional to r^2 — so a graph of area against r^2 is a straight line of gradient π.
- That doubling the radius makes the area four times larger, not twice.

Moving through it, one step at a time

Use the Circle to Parallelogram tool: slide the number of sectors up and watch the same slices rearrange into a parallelogram, then complete the statement and derive the formula yourself. Use the (r^2, A) Plotter: change the radius, plot G, and find the gradient and equation before revealing them. Then estimate the area of real objects. There is no rush: predict before you check, and use the symbol palette to enter π and the square.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (squaring a number, the area of a parallelogram, and the parts of a circle), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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